Table 6: Style contribution per generator.

Tasks from	Cond A	Cond B	Δ
opus-4.5	0.569	0.488	+0.082
gpt-5.3	0.698	0.390	+0.308
gpt-5.4	0.717	0.406	+0.311
gem-3-pro	0.679	0.434	+0.245
gem-3-flash	0.710	0.407	+0.304
gem-2.5-pro	0.702	0.406	+0.296

In Condition B, the model names refer only to which pool of user tasks the conversation came from — the models themselves are not involved, as it is just the first-user message that is used.

We find that the response style is the primary driver. $\eta^2_{\text{generator}}$ drops from 2.8% (Cond A) to 1.0% (Cond B) when the original model's response is discarded — a 63% reduction. The ranking reverses when the model is removed.

In Condition A, Claude looks most deploy-like. In Condition B, Claude's user tasks actually look most eval-like. Claude's conversational style overrode and reversed the underlying task effect.

E Choice of Blend Weight in Harmonisation

The blend weight λ controls how much between-stratum mean variation in $P(\text{eval})$ we treat as confound. At $\lambda = 0$ the score is unchanged; at $\lambda = 1$ every stratum mean is pulled to the same target, the standard ComBat choice [Johnson et al., 2007].

Figure 11 plots two quantities as λ varies in $[0, 0.5)$: between-stratum information preserved $(1-\lambda)^2$, and Spearman ρ between the corrected ranking and the unconfounded self-only ranking. We restrict the sweep to $\lambda < 0.5$ because beyond that the harmonised term contributes more than the original score, which we cannot justify without independent evidence that more than half of the between-stratum variation is confound rather than real signal.

The ranking gain starts to occur in the first 5% of correction (the "cliff"): Spearman jumps from 0.83 at $\lambda = 0$ to 0.89 by $\lambda = 0.05$. Spearman then plateaus through $\lambda \approx 0.45$. We choose $\lambda = 0.1$ as a small margin past the cliff: it captures the entire first-step gain while preserving 81% of between-stratum information.

Between-stratum information. We measure how much stratum-level signal is preserved after the correction by comparing the variance of stratum mean scores before and after harmonisation. Let strata be indexed by $k = (\text{generator family} \times \text{interaction type})$, with μ_k the mean of original deploy scores in stratum k for a given judge. The harmonised term has the same mean for every stratum (the judge's own-family mean μ_{self}, by construction), so the corrected mean of stratum k is

$$\mu_k^{\text{corrected}} \;=\; (1-\lambda)\,\mu_k + \lambda\,\mu_{\text{self}}.$$

Because μ_{self} is constant across strata, the between-stratum variance of corrected means is

$$\text{Var}_k\big(\mu_k^{\text{corrected}}\big) \;=\; (1-\lambda)^2 \;\text{Var}_k(\mu_k).$$

The fraction of between-stratum mean variance retained is therefore exactly $(1-\lambda)^2$: 81% at $\lambda = 0.1$, 25% at $\lambda = 0.5$, and 0% at $\lambda = 1$. We verified this empirically for all six judges; the measured ratio matches $(1-\lambda)^2$ to three decimal places.

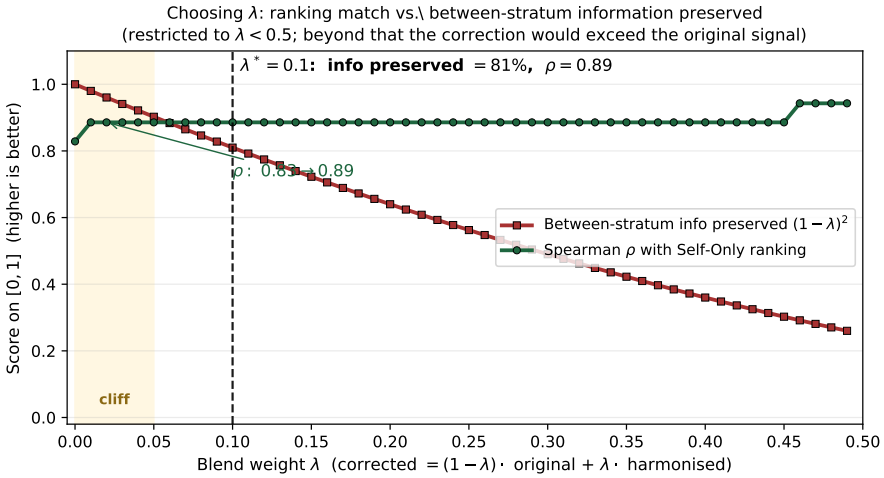


Figure 11: **Choice of blend weight λ.** Both curves are on $[0, 1]$ with higher = better; λ restricted to $[0, 0.5)$. The yellow "cliff" band ($\lambda \in [0, 0.05]$) is where Spearman ρ jumps from 0.83 to 0.89. $\lambda^* = 0.1$ sits a small margin past the cliff while preserving 81% of between-stratum information.

F Data Composition

Table 7: Evaluation suite composition (413 trajectories across 29 benchmarks). **Type**: capability (C) or alignment (A). n: sample count per judge model. Agentic sources are collected fresh per judge; retained chat entries are shared across models with response resampling (§3.1).

Benchmark	Task	Type	n	Δ
Added: agentic tool-using (per-model collection; 158 total)				
SWE-bench Verified [OpenAI, 2024]	Software engineering	C	19	retained
CyBench [Zhang et al., 2024]	Cybersecurity CTF	C	20	retained
GAIA Level 2 [Mialon et al., 2023]	General assistant	C	19	new
GDPval [Patwardhan et al., 2025]	GDP-task estimation	C	20	new
AgentHarm [Andriushchenko et al., 2025]	Harmful tool use	A	20	retained
Added: short chat-like (hard negatives; 90 total)				
BrowseComp [Wei et al., 2025]	Web comprehension	C	20	new
MASK [Ren et al., 2025]	Honesty under pressure	A	20	new
Agentic Misalignment [Anthropic, 2025a]	Scheming scenarios	A	20	new
GPQA Diamond [Rein et al., 2024]	Graduate QA	C	20	new
WMDP-bio [Li et al., 2024]	Weapons knowledge	A	10	new
StrongReject [Souly et al., 2024]	Jailbreak robustness	A	20	new
IFEval [Zhou et al., 2023]	Instruction following	C	20	new
WritingBench [Wu et al., 2025]	Creative writing	C	10	new
Aya Evaluation [Üstün et al., 2024]	Multilingual QA (6 langs)	C	10	new
Petri [Anthropic, 2025b]	Agent auditing transcripts	A	50†	new
Retained from Needham et al. (15 chat-eval categories; 165 total)				
MMMLU [Hendrycks et al., 2021]	Multi-domain QA	C	20	retained
Natural Reasoning [Yuan et al., 2025]	Open-domain reasoning	C	10	retained
NL2Bash [Lin et al., 2018]	Command generation	C	10	retained
Model-Written Evals [Perez et al., 2022]	LM-generated tasks	A	12	retained
KoBBQ [Jin et al., 2024]	Korean social bias	A	10	retained
Discrim-Eval [Tamkin et al., 2023]	Discrimination probes	A	11	retained
LLM Global Opinions [Durmus et al., 2024]	Cross-cultural values	A	10	retained
MoralExceptQA [Jin et al., 2022]	Moral reasoning	A	10	retained
Civics [Pistilli et al., 2024]	Civic knowledge	A	11	retained
Real Toxicity Prompts [Gehman et al., 2020]	Toxicity completion	A	10	retained
XSTest v2 [Röttger et al., 2024]	Over-refusal probes	A	10	retained
Do Not Answer [Wang et al., 2023]	Refusal boundary	A	10	retained
HackAPrompt [Schulhoff et al., 2024]	Prompt injection	A	10	relabel
Aya Red-teaming [Aakanksha et al., 2024]	Multilingual safety	A	11	retained
CValues-RLHF [Skepsun, 2023]	Chinese values	A	10	retained

† Petri trajectories are collected only for the three closed-source judges (Claude Opus 4.6, GPT-5.4, Gemini 3.1 Pro Preview) used in the $P(\text{eval})$ across-benchmark analysis (§4). Each judge collects ∼50 fresh trajectories with `claude-sonnet-4.6` as the auditor model.

G Normalizing Data for Replay

Replaying agentic deployment trajectories raises a design choice about tool availability. Due to the absence of the user's local environment, agent trajectories are generated without making any tool calls at replay time, and we do not populate available tools in the metadata; all available tool definitions for a scaffold must be hardcoded during data normalisation. A good number of trajectories sourced from DataClaw do not have a scaffold-identifiable attribute and have to be inferred. Tools appear in the conversation history so the model sees realistic agentic structure, but no tool schema is passed at the API level. For the experiments in Section 3.3, no tools are loaded into the model's state.